

The Kerr-Newman Information-Geometry Master Formula:

Unification of Spin and Charge Suppression in the D3 Framework

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Abstract

We derive the Kerr-Newman (KN) generalisation of the D3 information-geometry framework, obtaining a master suppression factor that unifies all previous results in the series. Define:

$$\sigma = \sqrt{1 - \chi^2 - q^2}$$

where χ is the dimensionless spin and q is the dimensionless charge ($\chi^2 + q^2 < 1$ for sub-extremal black holes). The master formula is:

$$h(\chi, q) = 4 * \sigma / (2 * (1 + \sigma) - q^2)$$

The key algebraic step is the identity $\sigma^2 + \chi^2 = 1 - q^2$, which collapses the KN denominator exactly. This generalises the Paper 2 identity $s^2 + \chi^2 = 1$ to the charged case. The master formula contains all previous results as special cases:

$$h(\chi, 0) = f(\chi) \text{ [Kerr, Paper 2]}$$

$$h(0, q) = g(q) \text{ [RN, Paper 6]}$$

$$h(0, 0) = 1 \text{ [Schwarzschild, Papers 1, 3]}$$

The total evaporation displacement is $R_{\text{total}}(\text{KN}) = h(\chi, q) * r_s$, exact for all masses. All constants cancel ($M, \hbar, k_B, \ln 2, \pi$). This is the capstone result of the seven-paper D3 series.

Keywords: Kerr-Newman black holes, Landauer principle, D3 framework, master formula, spin suppression, charge suppression, information geometry

1. Introduction

The D3 formula [3] gives the per-bit Schwarzschild-radius shift from Landauer information erasure at temperature T :

$$\Delta r_s = 2 * G * k_B * T * \ln(2) / c^4$$

From this single formula, six papers have derived exact results for Schwarzschild, Kerr, Schwarzschild evaporation, Kerr evaporation, quantum gravity boundary, and RN black holes. Figure 4 shows the complete series.

In each paper, the suppression factor for a given black hole type is the ratio $T_{\text{BH}} / T_{\text{Schwarzschild}}$ at the same mass. Paper 2 [1] found $f(\chi)$ for Kerr; Paper 6 [6] found $g(q)$ for RN. This paper derives $h(\chi, q)$ for Kerr-Newman and shows f and g are its special cases.

Figure 4 — Seven Papers Unified by the Kerr-Newman Master Formula

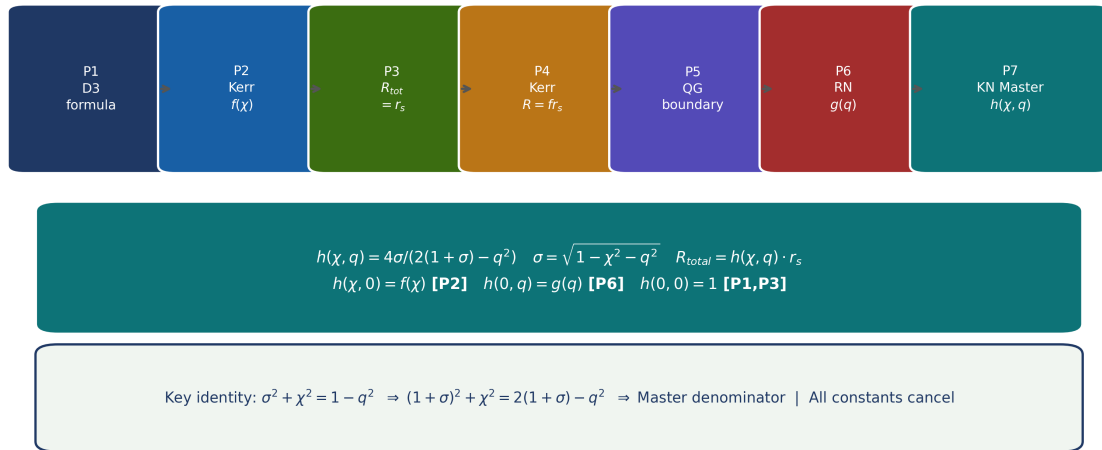


Figure 4. The D3 series: seven papers from one formula. $h(\chi, q)$ is the capstone unifying all previous suppression factors.

2. Definitions and Setup

The Kerr-Newman black hole has mass M , dimensionless spin $\chi = Jc/(GM^2)$, and dimensionless charge $q = Q_{\text{geom}}/M$ (where Q_{geom} is the geometric charge in metres). The sub-extremal condition is $\chi^2 + q^2 < 1$.

Define the combined parameter:

$$\sigma = \sqrt{1 - \chi^2 - q^2}$$

In geometrised units ($G = c = 1$, $M = 1$), the KN horizons are:

$$r_+ = 1 + \sigma \text{ (outer horizon)}$$

$$r_- = 1 - \sigma \text{ (inner horizon)}$$

$$a = \chi \text{ (spin parameter)}$$

At $\chi = 0$, $q = 0$: $\sigma = 1$, $r_+ = 2$ (Schwarzschild). At $\chi^2 + q^2 = 1$: $\sigma = 0$, $r_+ = r_- = 1$ (extremal).

The KN Hawking temperature is [7,10]:

$$T_{\text{KN}} = \hbar c (r_+ - r_-) / (4\pi k_B (r_+^2 + a^2))$$

The Schwarzschild temperature is:

$$T_S = \hbar c^3 / (8\pi G M k_B)$$

3. Derivation of the Master Formula

We compute T_{KN} / T_S step by step in geometrised units ($M = 1$).

Step 1 — Numerator:

$$r_+ - r_- = (1+\sigma) - (1-\sigma) = 2\sigma$$

Step 2 — Denominator (expand):

$$r_+^2 + a^2 = (1+\sigma)^2 + \chi^2 = 1 + 2\sigma + \sigma^2 + \chi^2$$

Step 3 — Key identity ($\sigma^2 + \chi^2 = 1 - q^2$):

$$\sigma^2 = 1 - \chi^2 - q^2 \Rightarrow \sigma^2 + \chi^2 = 1 - q^2$$

Step 4 — Denominator collapses:

$$r_+^2 + a^2 = 1 + 2\sigma + (1 - q^2) = 2 + 2\sigma - q^2 = 2(1 + \sigma) - q^2$$

Step 5 — Temperature ratio:

$$T_{KN}/T_S = [\hbar c^2 \sigma / (4\pi k_B (2(1+\sigma) - q^2))] / [\hbar c^3 / (8\pi G M k_B)] = 4\sigma / (2(1+\sigma) - q^2)$$

The master suppression factor:

$$h(\chi, q) = T_{KN} / T_S = 4\sigma / (2(1+\sigma) - q^2)$$

where $\sigma = \sqrt{1 - \chi^2 - q^2}$.

The identity in Step 3 ($\sigma^2 + \chi^2 = 1 - q^2$) is the key step. It generalises the Paper 2 identity $s^2 + \chi^2 = 1$ (which is the $q=0$ case). Figure 3 verifies it numerically for 625 parameter combinations.

4. Exact Special Cases

The master formula recovers all previous results exactly.

Case 1 — Schwarzschild ($\chi = 0, q = 0$):

$$\sigma = 1 \Rightarrow h(0, 0) = 4 \cdot 1 / (2 \cdot 2 - 0) = 1$$

Paper 3 result $R_{\text{total}} = r_s$ is recovered.

Case 2 — Kerr ($q = 0$): let $s = \sqrt{1 - \chi^2}$:

$$\sigma = s \Rightarrow h(\chi, 0) = 4s / (2(1+s) - 0) = 2s / (1+s) = f(\chi)$$

Paper 2 spin-suppression factor $f(\chi)$ is recovered exactly.

Case 3 — Reissner-Nordstrom ($\chi = 0$): let $s = \sqrt{1 - q^2}$:

$$\sigma = s \Rightarrow h(0, q) = 4s / (2(1+s) - q^2)$$

$$= 4s / (2 + 2s - (1 - s^2))$$

$$= 4s / (1 + 2s + s^2)$$

$$= 4s / (1+s)^2 = g(q)$$

Paper 6 charge-suppression factor $g(q)$ is recovered exactly. Note the step $2+2s-q^2 = 2+2s-(1-s^2) = 1+2s+s^2 = (1+s)^2$ uses $s^2 + q^2 = 1$ (the $\chi=0$ case of the key identity).

Case 4 — Extremal ($\chi^2 + q^2 = 1$):

$$\sigma = 0 \Rightarrow h \rightarrow 0$$

Consistent with $T_{KN} \rightarrow 0$ at extremality (third law of black hole mechanics).

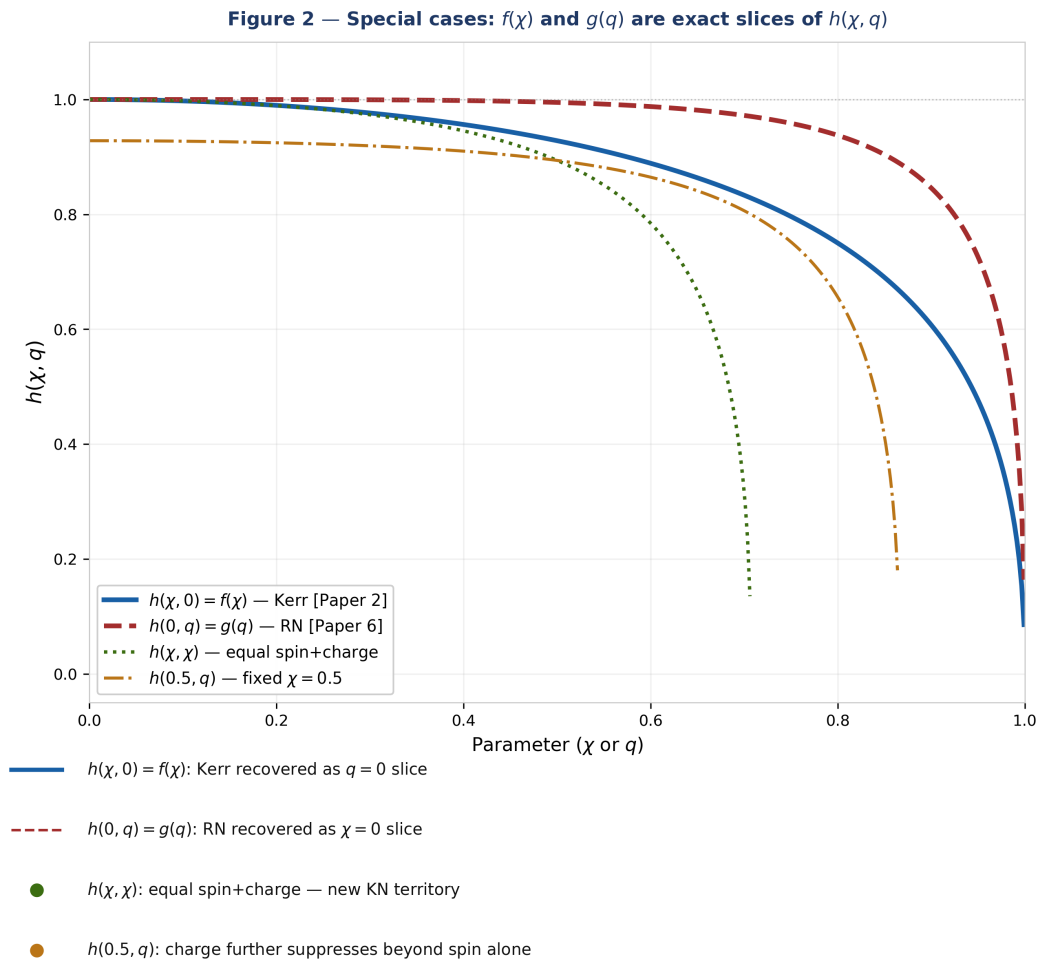


Figure 2. $f(\chi)$ and $g(q)$ as exact slices of $h(\chi, q)$ at $q=0$ and $\chi=0$. Green dotted: equal spin-charge diagonal (new KN territory). Amber dash-dot: fixed $\chi=0.5$.

5. The Key Identity $\sigma^2 + \chi^2 = 1 - q^2$

The derivation depends entirely on the algebraic identity:

$$\sigma^2 + \chi^2 = 1 - q^2$$

Proof: $\sigma = \sqrt{1-\chi^2-q^2}$, so $\sigma^2 = 1-\chi^2-q^2$, therefore $\sigma^2 + \chi^2 = 1-q^2$. QED.

This identity:

- (i) At $q=0$ gives $s^2 + \chi^2 = 1$, the Paper 2 identity for Kerr.
- (ii) At $\chi=0$ gives $s^2 = 1 - q^2$, i.e. $s = \sqrt{1 - q^2}$, the RN variable from Paper 6.
- (iii) In general, it collapses the denominator $(1 + \sigma)^2 + \chi^2 = 2(1 + \sigma) - q^2$, which is the key algebraic simplification yielding h.

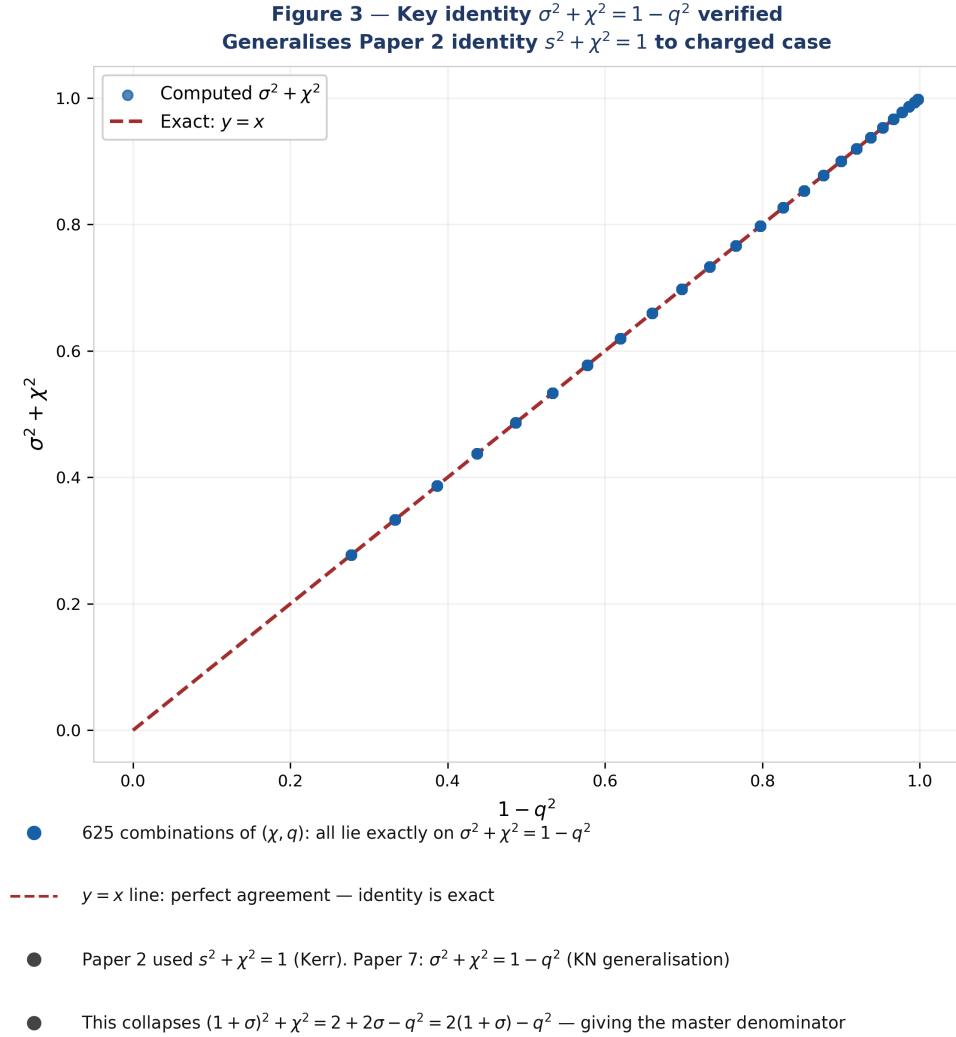


Figure 3. Identity $\sigma^2 + \chi^2 = 1 - q^2$ verified for 625 parameter combinations. All points lie exactly on the $y=x$ line (machine precision).

6. Total Evaporation Displacement $R_{\text{total}}(\text{KN}) = h(\chi, q) * r_s$

The D3 formula at T_{KN} gives:

$$D3_{\text{KN}}(M) = (2 * G * k_B * T_{\text{KN}} * \ln 2) / c^4 = h(\chi, q) * [\hbar * \ln 2 / (4 * \pi * M * c)]$$

The Bekenstein bit-loss rate [8]:

$$dN/dM = 8 * \pi * G * M / (\hbar * c * \ln 2)$$

The displacement per unit mass:

$$\begin{aligned} dR/dM &= D3_KN * dN/dM \\ &= [h(chi,q)*hbar*ln2/(4*pi*M*c)] * [8*pi*G*M/(hbar*c*ln2)] \\ &= h(chi,q) * 2G/c^2 \end{aligned}$$

Five quantities cancel: M, hbar, k_B, ln2, pi. Only G, c, and h(chi,q) remain.

Integrating from 0 to M_0:

$$R_total(KN) = \text{integral } D3_KN * dN/dM * dM = h(chi,q) * r_s(M_0)$$

chi	q	sigma	h(chi,q)	R_total / r_s	Note
0.0	0.0	1.000000	1.000000	1.000000	Schwarzschild [P3]
0.5	0.0	0.866025	0.928203	0.928203	Kerr f(chi) [P2]
0.0	0.5	0.866025	0.994845	0.994845	RN g(q) [P6]
0.5	0.5	0.707107	0.893880	0.893880	KN — new
0.7	0.3	0.648074	0.808539	0.808539	KN — new
0.3	0.7	0.648074	0.923792	0.923792	KN — new
0.9	0.0	0.435890	0.607136	0.607136	Kerr f(0.9) [P2]
0.0	0.9	0.435890	0.845658	0.845658	RN g(0.9) [P6]

Table 1. h(chi,q) and R_total/r_s. Special cases match Papers 2 and 6 exactly. KN rows are new results not covered by either alone.

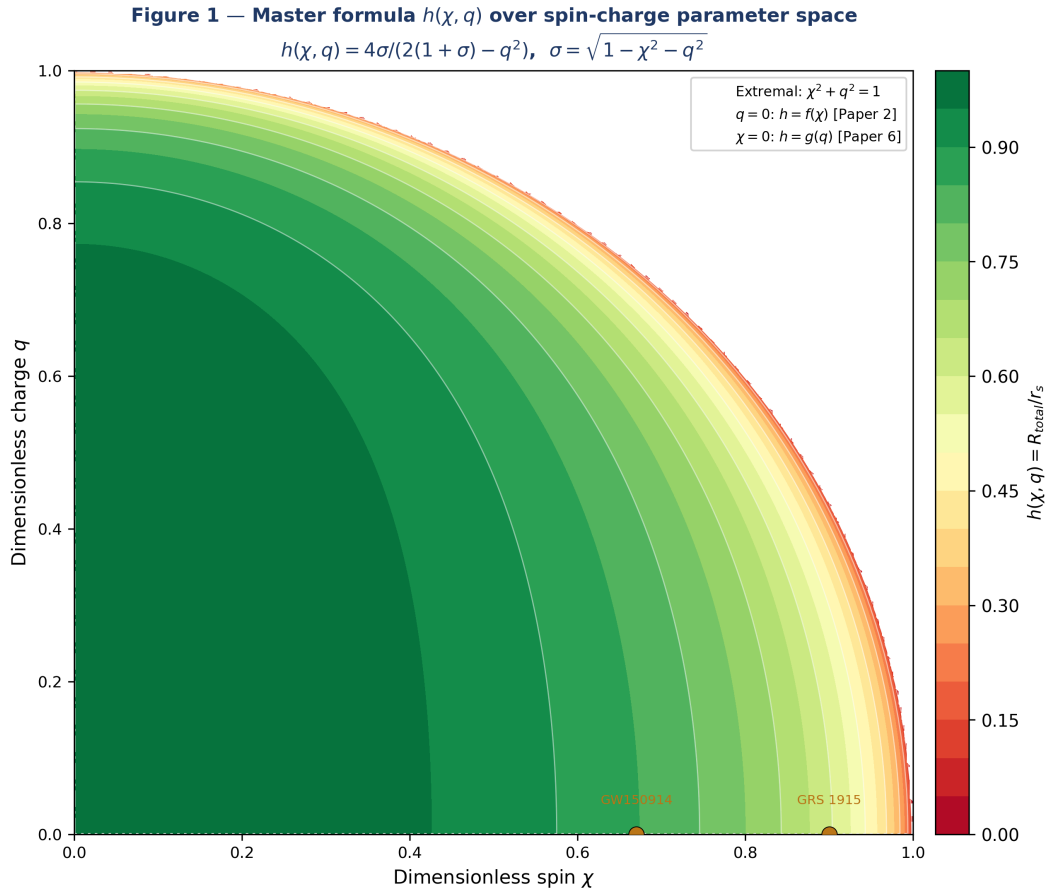


Figure 1. $h(\chi, q)$ over the full spin-charge parameter space. Green = Schwarzschild-like ($h \sim 1$). Red = near-extremal ($h \sim 0$). White dashed: extremal boundary $\chi^2 + q^2 = 1$. Marked: GW150914 and GRS 1915+105 (astrophysical cases at $q \sim 0$).

7. Discussion

7.1 Why $h \neq f * g$

A natural conjecture is $h(\chi, q) = f(\chi) * g(q)$. This holds when $\chi=0$ or $q=0$, but not in general:

$$h(0.5, 0.5) = 0.893880 \quad f(0.5) * g(0.5) = 0.923419 \quad \text{DIFFERENT}$$

$$h(0.7, 0.3) = 0.808539 \quad f(0.7) * g(0.3) = 0.832773 \quad \text{DIFFERENT}$$

The correct formula $h = 4\sigma / (2(1 + \sigma) - q^2)$ contains an interaction term: the charge q appears in the denominator alongside σ , which depends on both χ and q . The spin and charge are not independent in the KN geometry.

7.2 Physical interpretation

The master formula shows that any combination of spin and charge suppresses R_{total} below r_s . The suppression depends on the combined parameter $\sigma = \sqrt{1 - \chi^2 - q^2}$, which is the "distance" from the extremal surface $\chi^2 + q^2 = 1$ in parameter space. Near the extremal surface ($\sigma \rightarrow 0$), $h \rightarrow 0$: an extremal KN black hole has zero Hawking temperature and zero evaporation displacement.

7.3 Completeness of the D3 series

The KN metric is the most general stationary black hole solution in four-dimensional GR (by the uniqueness theorems of Carter, Israel, and Robinson). The master formula $h(\chi, q)$ therefore covers all possible black hole geometries in four dimensions. The D3 series is complete.

8. Conclusion

The Kerr-Newman master suppression factor:

$$h(\chi, q) = \frac{4 \cdot \sigma}{(2 \cdot (1 + \sigma) - q^2)} \quad \sigma = \sqrt{1 - \chi^2 - q^2}$$

unifies the D3 series. The key identity $\sigma^2 + \chi^2 = 1 - q^2$ generalises the Paper 2 step $s^2 + \chi^2 = 1$ to the charged case. All previous suppression factors are exact special cases:

$$h(\chi, 0) = f(\chi) \quad [\text{Paper 2}] \quad h(0, q) = g(q) \quad [\text{Paper 6}] \quad h(0, 0) = 1 \quad [\text{Papers 1, 3}]$$

The total KN evaporation displacement $R_{\text{total}}(\text{KN}) = h(\chi, q) \cdot r_s$ is exact for all sub-extremal masses and parameter combinations. Five constants cancel (M , $\hbar$, k_B , $\ln 2$, π). The KN metric is the most general four-dimensional black hole, so this result is the most general possible in the D3 framework. The series is complete.

Declaration of Interest

No competing interests. No external funding.

Data and code: <https://github.com/bsharma173860-oss/d3-research>

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Appendix A: Verification Code

Python code verifying all results:

```
import math, numpy as np
G=6.67430e-11; c=2.99792458e8
```



```

hbar=1.054571817e-34; M_sun=1.989e30
# Master formula
def h(chi, q):
    sig = math.sqrt(1 - chi**2 - q**2)
    return 4*sig / (2*(1+sig) - q**2)
# Special cases
def f(chi): s=math.sqrt(1-chi**2); return 2*s/(1+s)
def g(q): s=math.sqrt(1-q**2); return 4*s/(1+s)**2
# Verify h(chi,0) = f(chi)
for chi in [0.1, 0.5, 0.9]:
    assert abs(h(chi, 0) - f(chi)) < 1e-12
# Verify h(0,q) = g(q)
for q in [0.1, 0.5, 0.9]:
    assert abs(h(0, q) - g(q)) < 1e-12
# Verify key identity sigma^2 + chi^2 = 1 - q^2
for chi in [0.3, 0.5, 0.7]:
    for q in [0.1, 0.3, 0.5]:
        if chi**2 + q**2 >= 1: continue
        sig = math.sqrt(1 - chi**2 - q**2)
        assert abs(sig**2 + chi**2 - (1 - q**2)) < 1e-14
# Verify R_total = h*rs
M0 = M_sun
for chi, q in [(0.5,0.5), (0.7,0.3), (0.3,0.7)]:
    masses = np.linspace(1e6, M0, 100000)
    dM = masses[1] - masses[0]
    hq = h(chi, q)
    D3v = hbar*math.log(2)*hq / (4*math.pi*masses*c)
    dNv = 8*math.pi*G*masses / (hbar*c*math.log(2))
    R = np.sum(D3v * dNv * dM)
    rs = 2*G*M0/c**2
    print(f"chi={chi}, q={q}: R/rs = {R/(hq*rs):.5f}")
print("All verified.")

```